Explaining the Unseen: Multimodal Vision-Language Reasoning for Situational Awareness in Underground Mining Disasters

Mizanur Rahman Jewel, Mohamed Elmahallawy, Sanjay Madria, Samuel Frimpong
Missouri University of Science and Technology, Rolla, Missouri, USA

WACV 2026 TUCSON, AZ 3/6 – 3/10

Abstract

Underground mining disasters create darkness, smoke, debris, and collapse that obscure vision and make situational awareness difficult for responders. We present MDSE (Multimodal Disaster Situation Explainer), a vision-language framework that generates detailed, context-aware textual explanations of post-disaster underground scenes. Key ideas:

(i) segmentation-aware dual-pathway encoding that fuses global and region-specific embeddings,

(ii) context-aware cross-attention to align visual evidence with task cues

(iii) resource-efficient LoRA adaptation for caption generation with low training cost.

UMD dataset: web keyword search + visually realistic film scenes; 300 images with five captions each; standardized preprocessing (trim/normalize/224×224).

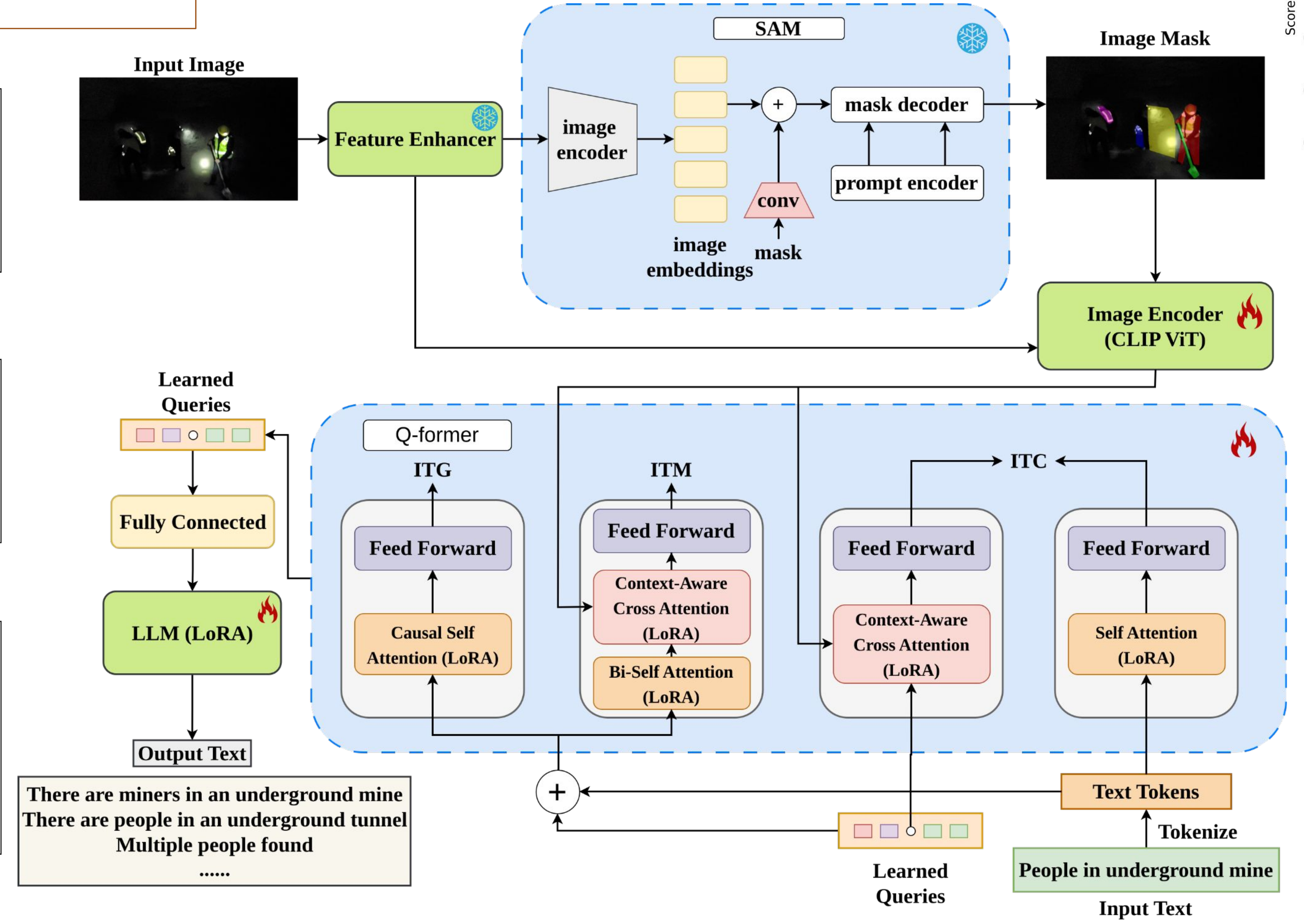
Annotation:
1. Disaster in an underground mine
2. Explosion in the tunnel
3. Equipment Explosion
4. Damaged pieces of equipment
5. Debris in the path of the tunnel

Annotation:
1. Fire Explosion
2. Fire hazard with fume
3. Explosion in the mine
4. Coal fume in the air
5. Fire in the mine entrance

Annotation:
1. Flood in the area
2. People stuck in the flood
3. Two people trying to escape the flood
4. Damaged properties because of the flood
5. Flood Disaster

Methods and Materials

- **Feature Enhancement (FE):** improves low-light/noisy images via multi-exposure fusion, tone mapping/contrast enhancement, white balance, and saturation correction.
- **Segmentation (SAM):** generates ROI masks even under poor visibility to isolate critical objects/regions.
- **Dual-pathway encoding (CLIP ViT):** encode enhanced image (global) and masks (region) to preserve both scene context and localized evidence; fuse via a learnable projection.
- **Context-aware Q-Former cross-attention:** inject external context (e.g., scene type / task cues) into attention keys to prioritize relevant visual evidence.
- **Efficient adaptation:** LoRA on Q-Former + LLM head



Results

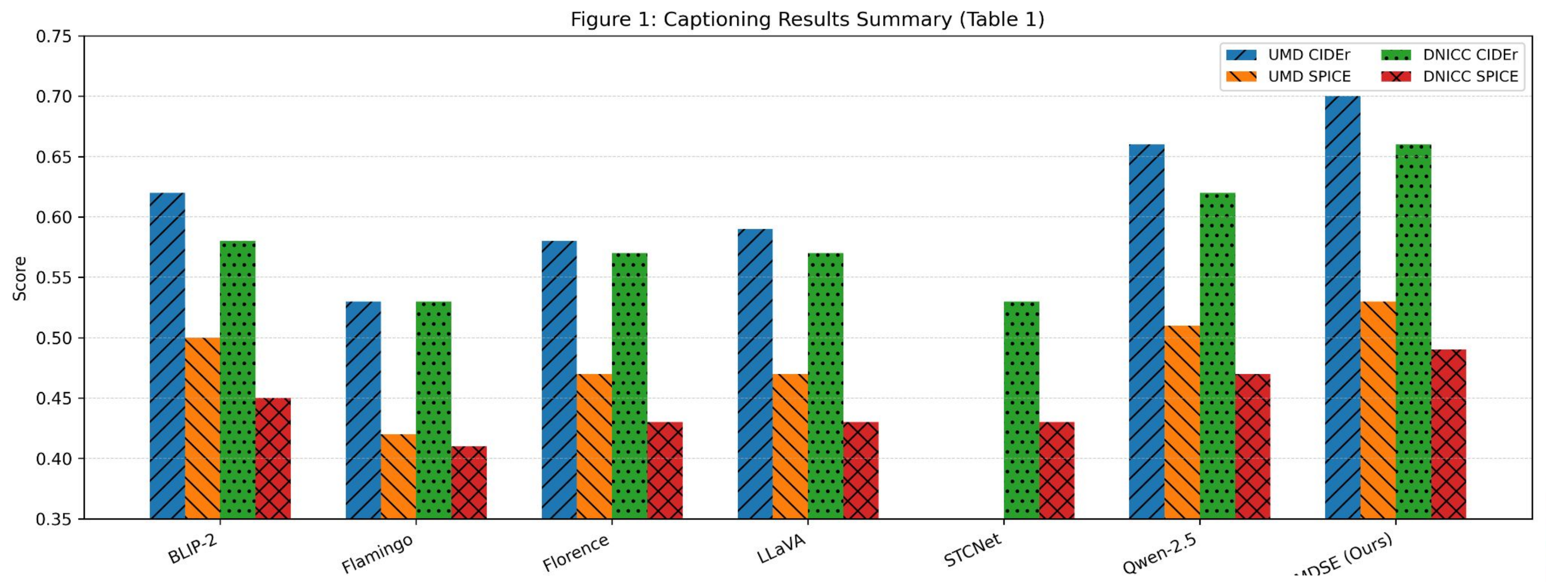


Figure 1: Captioning Results Summary (Table 1)

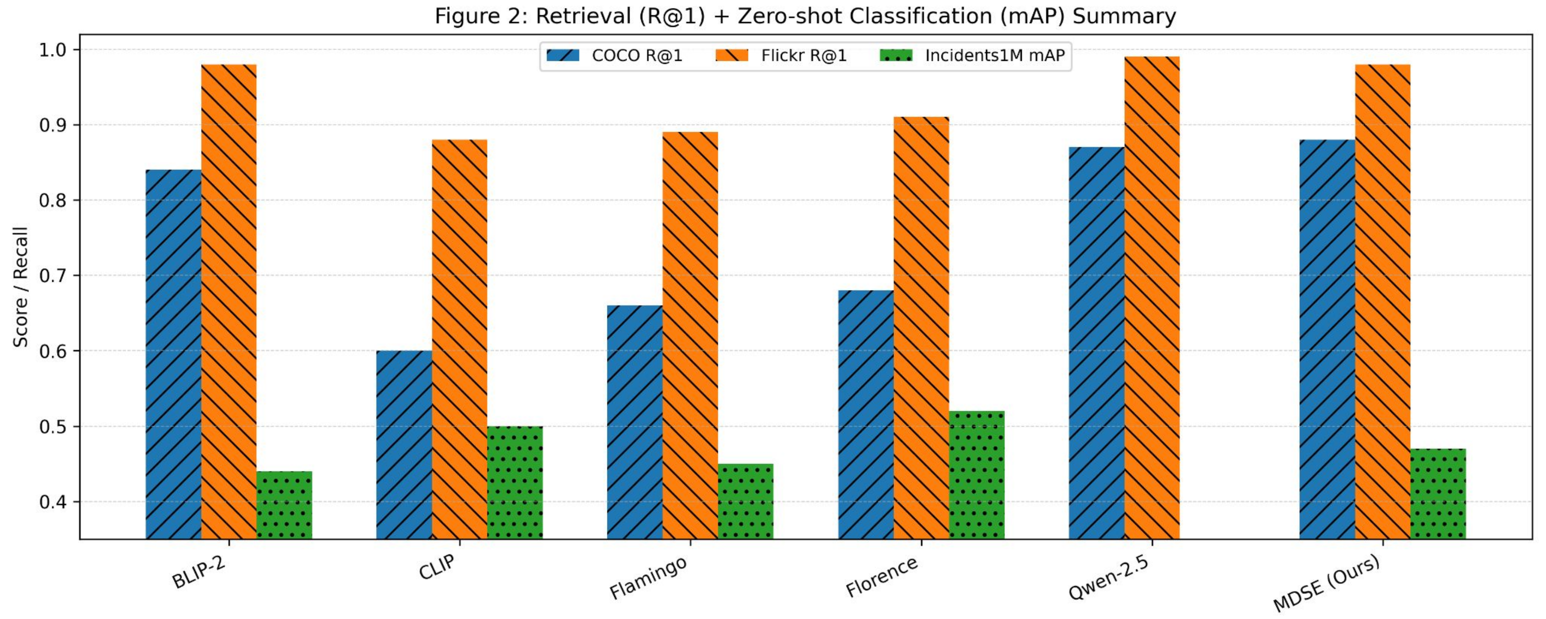


Figure 2: Retrieval (R@1) + Zero-shot Classification (mAP) Summary

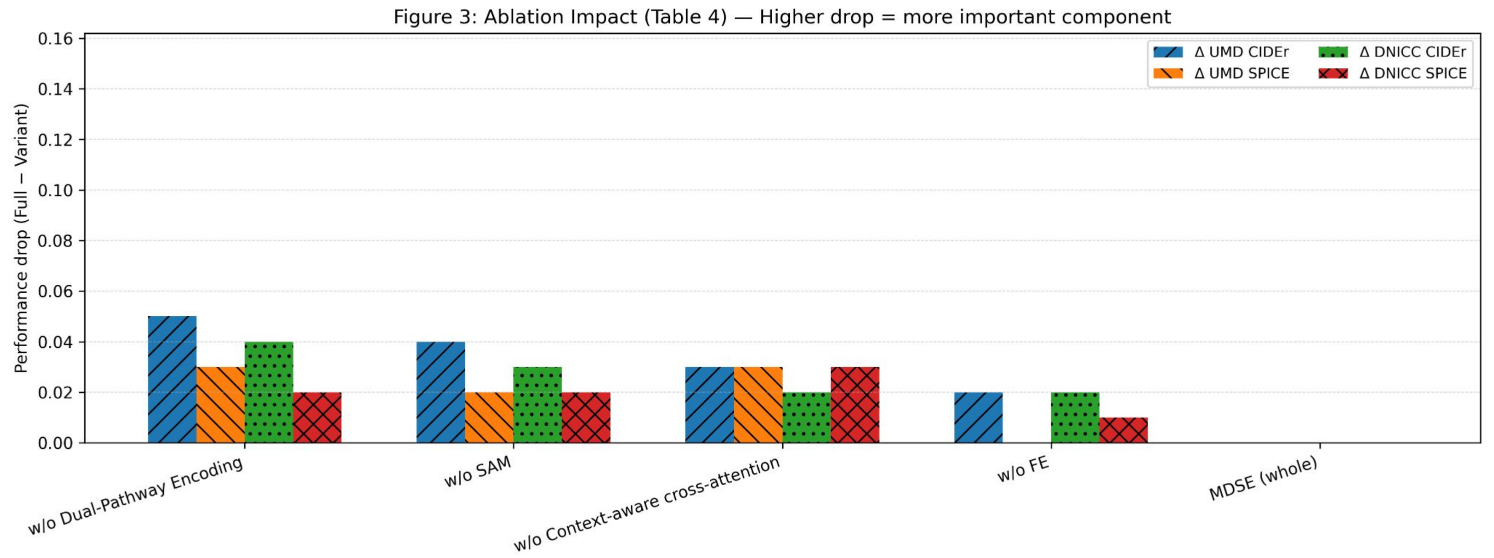


Figure 3: Ablation Impact (Table 4) — Higher drop = more important component

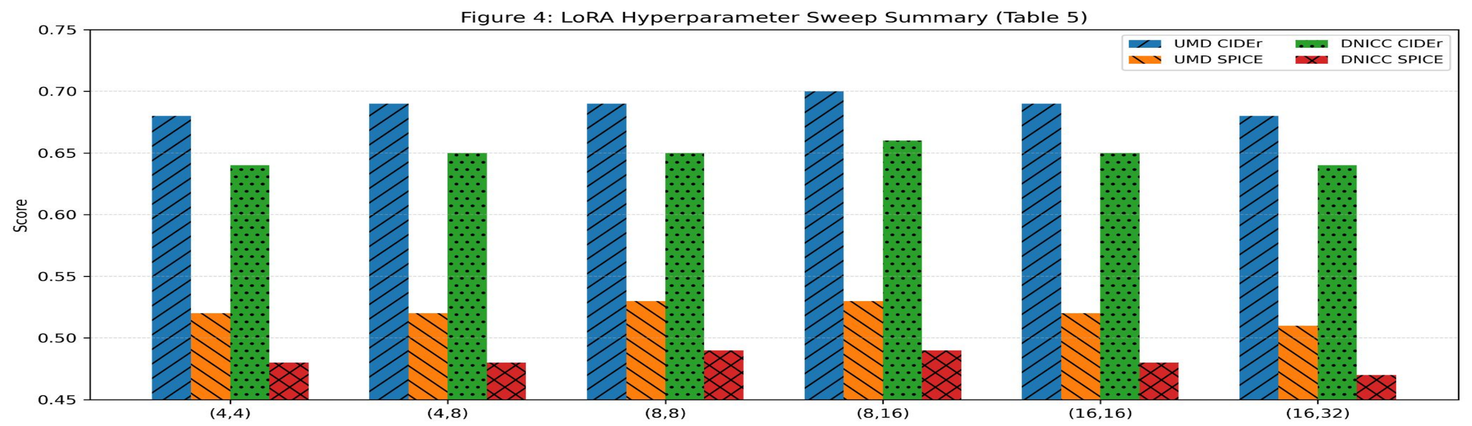


Figure 4: LoRA Hyperparameter Sweep Summary (Table 5)

Discussion

What MDSE adds in hard underground scenes:

- Uses SAM-guided regions + global context to retain subtle cues (e.g., blocked passages, structural damage) that are easy to miss in degraded visibility.
- Context-aware cross-attention improves relational/detail quality (reflected in SPICE).

A group of firefighters are working together to put out a fire.

A police officer stands in front of a car wreckage, observing the scene.

A car accident scene with a car on its side and another car in the background.

A large pile of debris is on fire, with smoke billowing into the sky.

A yellow truck driving through a tunnel

A man in a hard hat and helmet is working in a mine

A car is sitting in a tunnel with a light shining through it

A train is burning in a tunnel with smoke coming out

Conclusions

- **MDSE** generates detailed captions for underground disaster scenes using **segmentation-aware dual-path encoding** + **context-aware cross-attention**.
- Introduces **UMD**, a domain-specific underground disaster image–caption dataset (300 images, 5 captions/image).
- Achieves best captioning scores on **UMD** and **DNICC19k** among tested baselines.

Contact

Mizanur Rahman Jewel
Computer Science, Missouri S&T
Email: mj9vc@mst.edu
Website: mizanjewel.github.io
Phone: 5736472137

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